



Somantha Manuranga

Nationality: Sri Lankan **Date of birth:** 09/02/1997 **Place of birth:** Sri Lanka **Gender:** Male

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ABOUT ME

Master's student in AI for Industrial Applications with a background in Mechanical Engineering. Focused on Computer Vision, Model Robustness, and Digital Twins for Industry 4.0. Skilled in Python, PyTorch, Database systems and building production-ready AI pipelines. Seeking a Working Student position to solve complex automation and manufacturing challenges. Currently enhancing German language skills (B1 level).

EDUCATION AND TRAINING

Master of Artificial Intelligence for Industrial Applications

OTH Amberg-Weiden University, Germany [Current]

City: Amberg | Country: Germany | Website: www.oth-aw.de

- Focus on computer vision, NLP, autonomous robotics, and deep learning, machine learning, database systems.
- Applying machine learning to industrial processes.
- Modules: Computer Vision, Deep Learning, Autonomous Robotics, NLP

Bachelor of Science of Mechanical Engineering (Minor in Mechatronics)

University of Sri Jayewardenepura [19/11/2018 – 06/05/2024]

Thesis: Machine learning model for 2D aerodynamic drag prediction

Project: CoreXY 3D printer development integrating mechanical design and machine learning models.

WORK EXPERIENCE

🏢 KNSI Aviation Engineering

Design engineer

[08/08/2023 – 10/08/2024]

- Used SQL and Excel to manage compliance data workflows and technical documentation.
- Developed automated templates to reduce repetitive reporting effort.
- Contributed to documentation digitalization and gained insight into market-driven engineering processes.

🏢 Ceylon Biscuits Limited – Colombo, Sri Lanka

Trainee Mechatronic Engineer

[01/02/2022 – 15/07/2023]

- Contributed to process optimization through data-driven solutions and automation projects.
- Supported industry operations and collaborated with cross-functional teams to enhance productivity.
- Designed and fabricated 3D-printed automation components.

TECHNICAL SKILLS

Skills

- Python, C++, NoSQL, Database systems, MongoDB
- Machine Learning, NLP, Computer Vision, Huggingface
- PyTorch, Scikit-learn, OpenCV, TensorFlow (basic)
- NumPy, Pandas, Matplotlib, Seaborn
- MS Office Word, Excel, PowerPoint (confident user)
- LLM , VLM

PROJECTS

Projects & Research

Quantum Circuit Image Dataset & Classification Pipeline (BeautifulSoup, arXiv API)

- Built an automated pipeline to extract and classify 250+ quantum circuit images from arXiv “quant-ph” papers with linked JSON metadata.
- Fine-tuned a SciBERT-based caption classifier to detect quantum circuit figures, achieving an F1-score of 0.96.

Machine Learning Challenge - Winter 2025 (Edge AI)

- Developed ensemble models (XGBoost, LightGBM, CatBoost, stacking) for high-dimensional regression, increasing R-squared to 0.848.
- Created a simple 3-rule, 3-feature if-else model for edge deployment with R-squared ≈ 0.91.

ALIA Delivery – Hospital Digital Twin (Industry 4.0, Python, MQTT)

- Built a Webots-based digital twin of a hospital floor with a mobile robot delivering items to rooms based on web interface commands.
- Integrated an MQTT-based communication layer between a Python web application and the simulation, enabling room selection and autonomous robot navigation.

Bachelor’s Thesis - CNN-based Drag Prediction for 2D Geometries (Tensorflow)

- Developed convolutional neural networks to predict drag coefficients of arbitrary 2D shapes, replacing time-consuming CFD simulations.
- Trained on 12,000+ laminar-flow samples (Re = 10), achieving a minimum validation loss of 0.0017 and relative error below 1% for NACA airfoil profiles.
- Demonstrated real-time drag prediction suitable for aerodynamics applications (aircraft, automotive, civil structures) and AI-powered generative design workflows.

Social Media Addiction EDA - Student Dataset (Python)

- Conducted exploratory data analysis on 700+ student records to examine links between social media usage, academic performance, sleep quality, and mental health.
- Created publication-style visualizations (Matplotlib/Seaborn) and statistical summaries to highlight at-risk groups and support targeted intervention ideas.

Link: <https://github.com/SomanthaManuranga>

LANGUAGE SKILLS

Mother tongue(s): Sinhala

Other language(s):

English

LISTENING C1 READING C1 WRITING C1
SPOKEN PRODUCTION C1 SPOKEN INTERACTION C1

German

LISTENING B1 READING B1 WRITING B1
SPOKEN PRODUCTION B1 SPOKEN INTERACTION B1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

HOBBIES AND INTERESTS

Extra Curricular Activities & Leadership

- Team member in Football Club Amberg , Germany (FC Amberg)
- University Colors Award for Sports Excellence (2022)
- Football Captain - Engineering Faculty & University Team, Sri Lanka